

Highlights

PE-PINN: A Physics-Enhanced Physics-Informed Neural Network for Solving the PKN Model of Hydraulic Fracturing

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- A physics-enhanced PINN framework (PE-PINN) is proposed for the PKN model of hydraulic fracturing.
- A transient tip shape function analytically encodes the transition between asymptotic crack-tip regimes.
- A causal arrival-time constraint restricts PDE residual evaluation to the physically valid region.
- A learnable time-power exponent automatically discovers the correct temporal scaling behavior.

PE-PINN: A Physics-Enhanced Physics-Informed Neural Network for Solving the PKN Model of Hydraulic Fracturing[★]

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ARTICLE INFO

Keywords:

Physics-informed neural network

PKN model

hydraulic fracturing

Nordgren equation

moving boundary

tip singularity

ABSTRACT

This paper presents PE-PINN, a Physics-Enhanced Physics-Informed Neural Network framework for solving the PKN model of hydraulic fracturing. The governing Nordgren equation is solved by embedding physical priors directly into the network architecture, including a transient tip shape function that analytically interpolates asymptotic regimes, a causal arrival-time constraint, and a learnable time-power exponent. Numerical experiments demonstrate competitive accuracy against finite difference solutions while maintaining a mesh-free formulation.

1. Introduction

Hydraulic fracturing is a crucial technology for enhancing hydrocarbon recovery from unconventional reservoirs [Economides and Nolte \(2000\)](#). Accurate prediction of fracture geometry — the spatial and temporal evolution of fracture width, length, and net pressure — is essential for optimizing treatment design and evaluating well performance. Perkins and Kern and Nordgren first proposed the PKN model to describe the evolution of fracture length and width during hydraulic fracturing [Nordgren \(1972\)](#). This classical model, together with the KGD model [Geertsma and de Klerk \(1969\)](#), established the theoretical basis for the simulation of hydraulic fracturing. Among these models, the PKN model remains one of the most widely adopted in industrial practice due to its computational tractability. The PKN model assumes a fracture of fixed height that is much larger than its vertical extent, with plane-strain deformation in each cross-section. Under this assumption, the local elasticity relation yields a nonlinear governing equation that balances the temporal evolution of fracture width, the fluid leak-off into the surrounding formation, and the longitudinal gradient of the volumetric flow rate governed by the fluid's viscous resistance. Despite its apparent simplicity, this equation poses significant computational challenges due to its strong nonlinearity, moving free boundary, and singular behavior at the propagating tip.

The numerical solution of the PKN model has been an active research topic since Nordgren's original finite-difference formulation [Nordgren \(1972\)](#). Traditional grid-based approaches include explicit finite-difference schemes [Kemp \(1990\)](#), finite element methods with tip enrichment [Garikapati, Verhoosel, van Brummelen and Diez \(2017\)](#), and spectral discretizations [Peirce \(2014\)](#). However, these methods face persistent difficulties. The fracture tip exhibits a singular gradient that tends to infinity, requiring special tip elements, regularization techniques [Linkov \(2011\)](#), or asymptotic enrichment functions [Garikapati et al. \(2017\)](#) to avoid severe accuracy degradation. The classical PKN tip asymptote follows a one-third power-law in the viscosity-dominated regime [Nordgren \(1972\)](#); [Kemp \(1990\)](#), but a more complex transient behavior — smoothly transitioning from a one-third to a three-eighths power-law depending on the local leak-off intensity — has been recognized in recent re-examinations [Linkov \(2015\)](#); [Detournay \(2016\)](#). Furthermore, the moving free boundary evolves nonlinearly with time, following different power-law regimes depending on whether the propagation is dominated by fluid storage or leak-off. Accurately tracking this evolving crack front typically requires adaptive remeshing or front-tracking algorithms [Peirce \(2014\)](#), which introduce significant computational overhead. Additionally, a multi-scale structure arises from the thin boundary layer near the tip to

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the global fracture scale, demanding dense local grids that can make parametric studies and real-time applications prohibitive [Detournay \(2016\)](#).

Physics-Informed Neural Networks (PINNs) were introduced by Raissi *et al.* [Raissi, Perdikaris and Karniadakis \(2019\)](#) as a mesh-free framework that embeds the governing partial differential equations directly into the loss function of a neural network. PINNs exploit automatic differentiation for exact derivative computation and can naturally incorporate sparse observational data. Since their introduction, PINNs and their variants — including domain-decomposition extensions such as XPINN [Jagtap and Karniadakis \(2020\)](#) and cPINN [Jagtap, Kharazmi and Karniadakis \(2020\)](#), gradient-enhanced formulations [Yu, Lu, Meng and Karniadakis \(2022\)](#), and multi-scale architectures [Moseley, Markham and Nissen-Meyer \(2023\)](#) — have been successfully applied across a broad spectrum of problems in computational mechanics. Recently, PINNs have been applied to the field of subsurface engineering, including flow and transport in porous media [Hassan and Labak \(2024\)](#), history-matching in fractured reservoirs [Ali, Yan and Gunning \(2025\)](#), and two-phase flow in heterogeneous formations [Wang, Guo and Wu \(2024\)](#). However, the application of PINNs to hydraulic fracture propagation — characterized by its unique combination of a moving free boundary, singular tip asymptotics, and multi-regime transient behavior — remains relatively unexplored.

A small number of recent studies have begun to bridge PINNs with hydraulic fracture modeling. [Yang et al. \(2023\)](#) developed a physics-informed surrogate model for predicting hydraulic fracture geometry using deep learning, demonstrating the feasibility of replacing computationally expensive simulators with trained neural networks. [Tang et al. \(2024\)](#) proposed a PINN formulation with moving boundary constraints for modeling hydraulic fracturing, making an important step toward handling the free-boundary nature of the problem. However, existing PINN-based approaches for hydraulic fracturing still face critical limitations. First, most existing PINN formulations for fracture problems do not explicitly encode the asymptotic tip behavior into the network architecture. Without such prior knowledge, the network must learn sharp near-tip gradients purely from PDE residuals, which is notoriously difficult for gradient-based optimization [Krishnapriyan, Gholami, Zhe, Kirby and Mahoney \(2021\)](#). The transient transition between the one-third and three-eighths power-law tip asymptotics has not been addressed in any existing PINN work. Second, the PKN governing equation is physically valid only in the region where the fracture has already arrived. Existing PINN formulations do not enforce this causal constraint, leading to non-physical residuals being minimized in regions not yet reached by the fracture front. Third, the early-time behavior of fracture width exhibits a power-law singularity whose exponent depends on the dominant physical regime. Learning this temporal scaling directly from data is inefficient and limits generalization to time ranges not covered during training.

To address these challenges, this paper proposes PE-PINN (Physics-Enhanced PINN), a novel framework for the PKN model that explicitly embeds physical prior knowledge into the network architecture. The proposed method decomposes the fracture width into three physically meaningful components: a learnable time-power term that captures the early-time singular scaling, an analytically derived transient tip shape function that smoothly interpolates between the one-third and three-eighths asymptotic regimes, and a neural network that learns the remaining smooth component. This physics-enhanced representation explicitly separates the singular time and space dependencies, allowing the network to learn only a well-behaved residual function. Building on this decomposition, the arrival time of the crack tip at each spatial location is computed by numerically inverting the crack-length evolution relation, and the PDE residual evaluation is restricted to the physically meaningful region where the fracture has already arrived. This causal masking ensures physical consistency. Additionally, the time-power exponent is constrained to a physically plausible range and optimized jointly with the network parameters, automatically discovering the correct temporal scaling. A two-stage training strategy combining Adam [Kingma and Ba \(2015\)](#) for global exploration with L-BFGS [Liu and Nocedal \(1989\)](#) for fine local convergence is employed. In contrast to existing PINN formulations for hydraulic fracturing [Yang et al. \(2023\)](#); [Tang et al. \(2024\)](#), the proposed method encodes the asymptotic tip structure analytically rather than relying on the optimizer to discover it, which significantly reduces the learning burden on the network. Numerical experiments demonstrate that the proposed method achieves competitive accuracy against conventional finite difference solutions while maintaining a mesh-free formulation. Systematic ablation studies and sensitivity analyses further confirm that each component of the proposed framework contributes measurably to overall accuracy.

The remainder of this paper is organized as follows. Section 2 formulates the PKN governing equations. Section 3 describes the proposed PE-PINN framework, including the network design, loss function, training strategy, and experimental configuration. Section 4 presents the numerical results. Section 5 concludes the paper.

2. Problem Formulation

Consider a vertical hydraulic fracture of constant height H propagating within a homogeneous, isotropic, and linear elastic reservoir. A Newtonian fracturing fluid is injected at a constant volumetric rate Q_0 at the wellbore, driving the fracture to extend perpendicular to the minimum in-situ stress. The following assumptions are adopted Nordgren (1972); Detournay (2016): (i) the fracture length $L(t)$ is much larger than its height ($L \gg H$), and the height is much larger than the maximum opening ($H \gg W$), so that plane-strain deformation prevails in each vertical cross-section; (ii) the fluid flow within the narrow fracture channel is laminar and governed by lubrication theory; (iii) fluid leak-off from the fracture faces into the surrounding permeable formation follows Carter's one-dimensional leak-off law, under which the leak-off velocity at a given location decays inversely with the square root of the elapsed time since the fracture tip first reached that location.

Under plane-strain conditions, the net pressure $p(x, t)$ — the difference between the fluid pressure inside the fracture and the far-field confining stress — is linearly proportional to the local maximum fracture opening $W(x, t)$. For a fracture of elliptical vertical cross-section, this relation takes the form Nordgren (1972)

$$p(x, t) = \frac{E'}{2H} W(x, t), \quad (1)$$

where $E' = E/(1-\nu^2)$ is the plane-strain Young's modulus, E is the Young's modulus, and ν is the Poisson's ratio. The volumetric flow rate $q(x, t)$ within the fracture is governed by Poiseuille flow in a narrow elliptical conduit. Exploiting $W \ll H$, the flow rate relates to the longitudinal gradient of W^4 as Nordgren (1972); Kemp (1990)

$$q(x, t) = -\frac{\pi E'}{512\mu} \frac{\partial}{\partial x} [W^4(x, t)], \quad (2)$$

where μ is the dynamic viscosity of the fracturing fluid.

Local mass conservation balances the temporal change in the elliptical cross-sectional area $A = (\pi/4)WH$, the longitudinal flux gradient, and the fluid volume lost to the formation through both fracture faces:

$$\frac{\pi H}{4} \frac{\partial W}{\partial t} + \frac{\partial q}{\partial x} + \frac{2C_L H}{\sqrt{t - \tau(x)}} = 0, \quad (3)$$

where C_L is the Carter leak-off coefficient and $\tau(x)$ is the arrival time — the instant at which the propagating fracture tip first reaches the spatial coordinate x .

Substituting Eqs. (2) and (1) into Eq. (3) yields the Nordgren equation, a nonlinear diffusion-type PDE governing the spatiotemporal evolution of the fracture width:

$$\frac{\partial W}{\partial t} + \frac{8C_L}{\pi \sqrt{t - \tau(x)}} - \frac{E'}{128\mu H} \frac{\partial^2}{\partial x^2} (W^4) = 0. \quad (4)$$

The initial and boundary conditions are

$$W(x, 0) = 0, \quad L(0) = 0, \quad (5)$$

$$\left. \frac{\partial (W^4)}{\partial x} \right|_{x=0} = -\frac{256\mu Q_0}{\pi E'}, \quad W(L(t), t) = 0, \quad (6)$$

where the inlet condition follows from $q(0, t) = Q_0/2$. The arrival time $\tau(x)$ is defined implicitly by $L(\tau) = x$, ensuring that the leak-off term in Eq. (4) is evaluated only after the fracture has reached a given location.

To expose the self-similar structure of the problem and eliminate dimensional parameters, characteristic scales W_* , L_* , and T_* are introduced such that the dimensionless diffusion and leak-off coefficients equal unity and the dimensionless inlet flux equals one. Defining

$$\Omega = \frac{W}{W_*}, \quad \xi = \frac{x}{L_*}, \quad \bar{t} = \frac{t}{T_*}, \quad \bar{\tau} = \frac{\tau}{T_*}, \quad (7)$$

Eq. (4) reduces to the compact dimensionless form [Kemp \(1990\)](#); [Detournay \(2016\)](#)

$$\frac{\partial \Omega}{\partial \bar{t}} + \frac{1}{\sqrt{\bar{t} - \bar{\tau}(\xi)}} - \frac{\partial^2}{\partial \xi^2} (\Omega^4) = 0, \quad (8)$$

with the normalized boundary condition

$$\left. \frac{\partial (\Omega^4)}{\partial \xi} \right|_{\xi=0} = -1, \quad \Omega(\bar{L}(\bar{t}), \bar{t}) = 0, \quad (9)$$

where $\bar{L}(\bar{t}) = L(t)/L_*$. Hereafter, the bars on dimensionless quantities are omitted for brevity; all symbols $\{x, t, \tau, W, L\}$ denote the dimensionless variables unless otherwise stated.

The fracture length $L(t)$ follows distinct power-law regimes depending on the dominant physical mechanism. In the storage-dominated regime, where most of the injected fluid remains within the fracture, the tip propagation obeys $L(t) \propto t^{4/5}$. In the leak-off-dominated regime, where fluid loss governs the volume balance, the propagation slows to $L(t) \propto t^{1/2}$ [Nordgren \(1972\)](#); [Kemp \(1990\)](#). Between these two limits, the length evolution is well approximated by a harmonic mean interpolation of the asymptotic solutions [Detournay \(2016\)](#):

$$L(t) = \left[\frac{1}{\gamma_1^2 t^{8/5}} + \frac{1}{\gamma_2^2 t} \right]^{-1/2}, \quad (10)$$

where $\gamma_1 \approx 1.3514$ and $\gamma_2 \approx 0.6366$ are dimensionless constants determined by the self-similar solutions of the storage-dominated and leak-off-dominated regimes, respectively [Kemp \(1990\)](#); [Detournay \(2016\)](#).

A defining feature of the PKN model is the singular near-tip structure of the fracture width. As $x \rightarrow L(t)$ from below, the width vanishes but its longitudinal gradient becomes unbounded. Defining the dimensionless distance from the tip as $\zeta = 1 - x/L(t)$, the near-tip width follows a one-third power law ($W \propto \zeta^{1/3}$) in the storage-dominated regime and a three-eighths power law ($W \propto \zeta^{3/8}$) in the leak-off-dominated regime [Linkov \(2015\)](#); [Detournay \(2016\)](#). In the general transient case, where both fluid storage and leak-off contribute, the tip shape transitions smoothly between the two limiting asymptotes. An analytical expression capturing this transition was derived by [Linkov \(2015\)](#):

$$\Omega_{\text{tip}}(\zeta) = \omega_0(\zeta) \cdot 3^{1/3} \zeta^{1/3}, \quad (11)$$

where $\omega_0(\zeta)$ is a smooth modulation function satisfying $\omega_0 \rightarrow 1$ as $\zeta \rightarrow 0$ (recovering the classical one-third behavior at the very tip) and $\omega_0 \propto \zeta^{1/24}$ at larger ζ (yielding the three-eighths behavior farther from the tip).

In summary, Eqs. (8), (5), (9), (10), and (11) form a fully coupled system that governs the spatiotemporal evolution of the fracture width. This system constitutes the forward problem that the PE-PINN framework developed in Section 3 is designed to solve.

3. Methodology

We present the PE-PINN framework in three parts: the physics-enhanced network architecture (Section 3.1), the loss formulation and training algorithm (Section 3.2), and the experimental configuration (Section 3.3).

3.1. PE-PINN design

We propose a physics-enhanced representation that factorizes the fracture width into three components:

$$\widehat{W}(x, t; \theta, \alpha) = t^\alpha \Omega_{\text{tip}} \left(1 - \frac{x}{L(t)} \right) \mathcal{N}_\theta(x, t), \quad (12)$$

where $\mathcal{N}_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^+$ is a fully connected neural network with parameters θ , $\Omega_{\text{tip}} : [0, 1] \rightarrow [0, 1]$ is the transient tip shape function defined in Eq. (11), and $\alpha \in [0.1, 0.3]$ is a learnable time-power exponent. The factorization (12) is designed so that the singular temporal and spatial dependencies are handled analytically, leaving $\mathcal{N}_\theta$ to represent only a smooth, well-behaved residual.

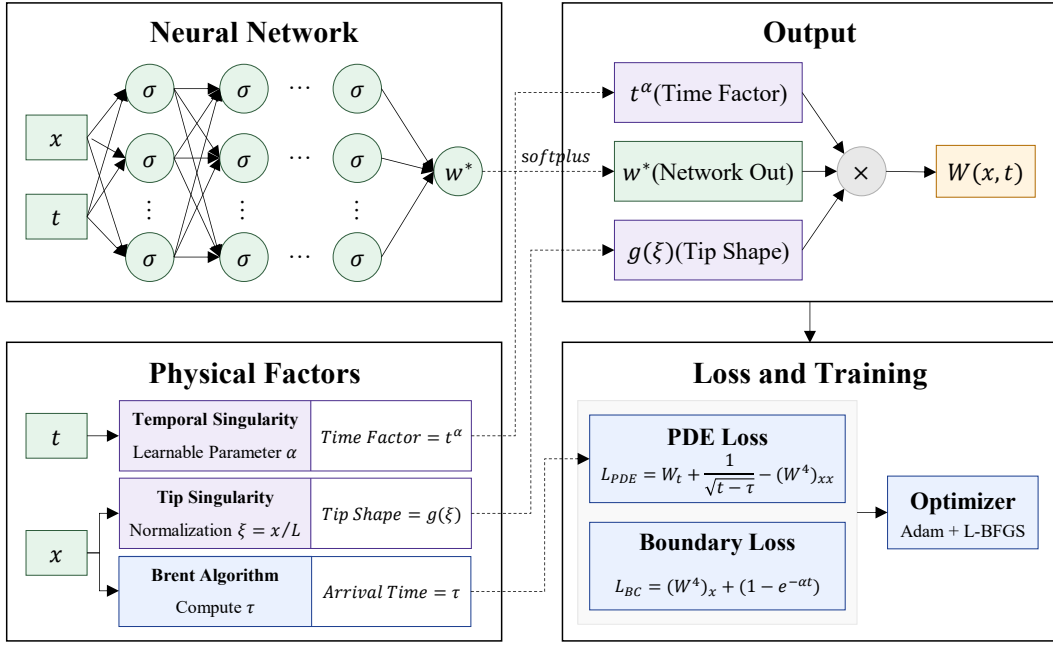


Figure 1: Schematic of the PE-PINN architecture. The network input consists of the dimensionless spatial and temporal coordinates (x, t) . The output is a positive scalar field combined with the time-power factor t^α and the transient tip shape function $\Omega_{\text{tip}}(\xi)$ to produce the fracture width $\widehat{W}(x, t)$. The PDE residual and boundary condition are embedded in the loss function, and the network is trained using a two-stage Adam + L-BFGS strategy.

The architecture of $\mathcal{N}_\theta$, illustrated in Fig. 1, is a fully connected network with four hidden layers of 128 neurons each and the hyperbolic tangent (Tanh) activation. The output layer consists of a single neuron followed by the Softplus function $\sigma(z) = \ln(1 + e^z)$, ensuring positivity. Let $\theta = \{W^{(\ell)}, b^{(\ell)}\}_{\ell=1}^L$ denote the collection of weights and biases. The weights are initialized via Xavier normal sampling $W_{ij}^{(\ell)} \sim \mathcal{N}(0, 2/(n_{\ell-1} + n_\ell))$ and biases are initialized to zero.

The tip shape function $\Omega_{\text{tip}}(\xi)$ encodes the asymptotic structure near the propagating front. Define the normalized tip coordinate $\xi = 1 - x/L(t) \in [0, 1]$. The function satisfies the two-scale asymptotic matching condition

$$\Omega_{\text{tip}}(\xi) \sim \begin{cases} \xi^{1/3}, & \xi \rightarrow 0 \quad (\text{storage-dominated}), \\ \xi^{3/8}, & \xi \rightarrow 1 \quad (\text{leak-off-dominated}), \end{cases} \quad (13)$$

with an explicit closed-form expression given by Eq. (11) that provides a smooth interpolation between the two limiting regimes. By embedding this asymptotic structure directly in the ansatz, the network is relieved of the burden of learning the singular near-tip gradient.

A further structural constraint arises from the domain of validity of the Nordgren equation. Equation (8) governs the fracture width only within the domain

$$\mathcal{V} = \{(x, t) \in \mathbb{R}^+ \times \mathbb{R}^+ \mid x \leq L(t), t > \tau(x)\}, \quad (14)$$

where $\tau(x)$ is the arrival time at which the crack tip first reaches position x . Evaluating the PDE residual outside $\mathcal{V}$ introduces non-physical contributions. The arrival time is defined implicitly through the crack-length evolution:

$$\tau(x) = \{t > 0 \mid L(t) = x\}, \quad (15)$$

which is solved numerically at each collocation point via Brent's root-finding method applied to Eq. (10). With $\tau(x)$ precomputed, a binary validity mask is constructed and the PDE residual is evaluated exclusively on $\mathcal{V}$.

The exponent α in (12) is treated as a learnable parameter. It is initialized at $\alpha_0 = 0.20$ and, after each gradient step, projected onto the admissible interval

$$\alpha \in \mathcal{A} = [0.1, 0.3], \quad (16)$$

via hard clipping: $\alpha \leftarrow \max(0.1, \min(0.3, \alpha))$. This parameter is optimized jointly with θ under the total loss (17). Physically, the optimal α reflects the dominant propagation regime: $\alpha = 0.25$ for strictly storage-dominated propagation, with lower values indicating a transition toward leak-off control.

The factorization (12) possesses two structural properties that simplify the loss formulation.

- P1. Automatic satisfaction of the initial condition.** At $t = 0$, the time-power factor vanishes: $t^\alpha|_{t=0} = 0$ for any $\alpha > 0$, hence $\widehat{W}(x, 0) = 0$, matching Eq. (5).
- P2. Automatic satisfaction of the right boundary condition.** At the crack tip $x = L(t)$, the tip coordinate becomes $\zeta = 1 - L(t)/L(t) = 0$. Since $\Omega_{\text{tip}}(0) = 0$ by construction, it follows that $\widehat{W}(L(t), t) = 0$, satisfying $\Omega(\bar{L}(\bar{t}), \bar{t}) = 0$ in Eq. (9).

Consequently, the learning problem reduces to enforcing the PDE residual (8) in the interior domain $\mathcal{V}$ and the inlet flux condition at $x = 0$. No separate loss terms for the initial condition or the tip condition are required.

3.2. Loss function and training

The proposed framework is trained without observational data. Let θ denote the network parameters and α the learnable time exponent. The training objective is constructed from the governing PDE and the inlet boundary condition alone:

$$\min_{\theta, \alpha \in \mathcal{A}} \mathcal{L}_{\text{total}}(\theta, \alpha) = \mathcal{L}_{\text{PDE}}(\theta, \alpha) + \mathcal{L}_{\text{BC}}(\theta, \alpha), \quad (17)$$

where $\mathcal{A} = [0.1, 0.3]$ is the admissible range for α . No weighting coefficients are introduced; the dimensionless scaling of (8) ensures that both terms are naturally of comparable magnitude.

PDE residual. Substituting the ansatz (12) into the dimensionless Nordgren equation (8) yields the pointwise residual

$$\mathcal{R}_{\text{PDE}}(x, t) = \frac{\partial \widehat{W}}{\partial t} + \frac{1}{\sqrt{t - \tau(x)}} - \frac{\partial^2}{\partial x^2} (\widehat{W}^4), \quad (18)$$

where $\partial \widehat{W} / \partial t$ and $\partial^2 (\widehat{W}^4) / \partial x^2$ are evaluated exactly via automatic differentiation. The PDE loss aggregates this residual over the causal domain $\mathcal{V}$ defined in (14):

$$\mathcal{L}_{\text{PDE}}(\theta, \alpha) = \frac{1}{|\mathcal{D}_{\text{PDE}}|} \sum_{(x_i, t_i) \in \mathcal{D}_{\text{PDE}}} \mathbf{1}_{\mathcal{V}}(x_i, t_i) [\mathcal{R}_{\text{PDE}}(x_i, t_i)]^2, \quad (19)$$

where $\mathcal{D}_{\text{PDE}}$ is the set of PDE collocation points, $\mathbf{1}_{\mathcal{V}}$ is the indicator function of the valid domain, and $|\mathcal{D}_{\text{PDE}}|$ denotes cardinality.

Boundary condition loss. The inlet condition (9) imposes $\partial(\widehat{W}^4) / \partial x = -1$ at $x = 0$ for $t > 0$. However, at $(x, t) = (0, 0)$, the initial condition $\widehat{W}(x, 0) = 0$ and this flux condition are incompatible — the former requires zero flux while the latter demands a finite injection. This incompatibility is a standard feature of moving boundary problems with a starting injection. We introduce a smooth ramp function

$$S(t) = 1 - e^{-\beta t}, \quad \beta = 100, \quad (20)$$

which satisfies $S(0) = 0$ and $S(t) \approx 1$ for $t > 0.05$, thereby suppressing the boundary constraint at the temporal origin. The boundary loss is then

$$\mathcal{L}_{\text{BC}}(\theta, \alpha) = \frac{1}{|\mathcal{D}_{\text{BC}}|} \sum_{t_i \in \mathcal{D}_{\text{BC}}} \left[\left. \frac{\partial(\widehat{W}^4)}{\partial x} \right|_{(0, t_i)} + S(t_i) \right]^2, \quad (21)$$

where $\mathcal{D}_{\text{BC}}$ is a set of time points drawn from $[0, T_{\text{max}}]$ at $x = 0$.

Collocation strategy. Collocation points are drawn via Latin Hypercube Sampling (LHS) to achieve space-filling coverage of the spatiotemporal domain. Three strata are defined. Define the dimensionless time partitions $\mathcal{T}_1 = [0, 10]$

(early) and $\mathcal{T}_2 = [10, 80]$ (late), and the near-tip stratum $\mathcal{S}_{\text{tip}} = \{(x, t) \mid x/L(t) \in [0.95, 0.999]\}$. The PDE collocation set is

$$\mathcal{D}_{\text{PDE}} = \mathcal{D}_{\text{PDE}}^{(1)} \cup \mathcal{D}_{\text{PDE}}^{(2)} \cup \mathcal{D}_{\text{PDE}}^{\text{tip}}, \quad (22)$$

with $|\mathcal{D}_{\text{PDE}}^{(1)}| = 10,000$, $|\mathcal{D}_{\text{PDE}}^{(2)}| = 20,000$, and $|\mathcal{D}_{\text{PDE}}^{\text{tip}}| = 10,000$ (total 40,000 PDE points). Spatial coordinates are mapped to the physical domain via $x = \xi \cdot L(t)$, where $\xi \sim \mathcal{U}(0, 1)$ for the first two strata. The boundary set $\mathcal{D}_{\text{BC}}$ comprises 2,000 points at $x = 0$ with $t \sim \mathcal{U}(0, 80)$. The arrival time $\tau(x)$ is precomputed via (15) for every PDE point; the indicator $\mathbf{1}_{\mathcal{V}}$ is evaluated once and held fixed throughout training.

Two-stage optimization. The loss landscape of (17) is highly non-convex due to the nonlinear PDE constraint and the multiplicative coupling between $\mathcal{N}_{\theta}$ and the physics-enhanced factors. We adopt a two-stage strategy that leverages the complementary strengths of first-order stochastic descent and second-order quasi-Newton refinement.

Stage 1 — Adam. The Adam optimizer Kingma and Ba (2015) is applied for $K_1 = 5,000$ iterations with initial learning rate $\eta_0 = 10^{-3}$, exponential decay $\eta_k = \eta_0 \cdot \gamma^k$ with $\gamma = 0.9995$, and gradient clipping at $\|\nabla_{(\theta, \alpha)} \mathcal{L}\|_{\max} = 1.0$. This stage provides robust global exploration and prevents premature convergence to poor local minima.

Stage 2 — L-BFGS. The L-BFGS optimizer Liu and Nocedal (1989) with strong Wolfe line search is initialized from the Adam-converged parameters for up to $K_2 = 10,000$ iterations. The optimization uses a history size $m = 100$, learning rate $\eta = 0.5$, and convergence tolerances $\varepsilon_g = 10^{-6}$ (gradient norm) and $\varepsilon_p = 10^{-9}$ (parameter change). Early stopping is activated when $|\mathcal{L}^{(k)} - \mathcal{L}^{(k-1)}| < 10^{-10}$ for 200 consecutive iterations. The complete procedure is formalized in Algorithm 1.

Algorithm 1 Two-stage training procedure for PE-PINN.

- 1: **Input:** Physical parameters $\{E', H, \mu, C_L, Q_0\}$, domain $[0, T_{\max}]$
 - 2: **Output:** Trained network parameters θ , learnable exponent α
 - 3: **// — Collocation and precomputation —**
 - 4: Sample PDE and boundary collocation points via LHS
 - 5: Map spatial coordinates: $x = \xi \cdot L(t)$
 - 6: Precompute arrival time $\tau(x_i)$ at each PDE point via Brent's method (Eq. 15)
 - 7: **// — Stage 1: Adam pre-training —**
 - 8: Initialize network weights θ with Xavier scheme, $\alpha \leftarrow 0.20$
 - 9: **for** $k = 1$ to K_1 **do**
 - 10: Compute $\mathcal{L} \leftarrow \mathcal{L}_{\text{PDE}} + \mathcal{L}_{\text{BC}}$
 - 11: Update θ, α via Adam with gradient clipping
 - 12: **end for**
 - 13: **// — Stage 2: L-BFGS fine-tuning —**
 - 14: Load Adam-converged parameters
 - 15: **for** $k = 1$ to K_2 **do**
 - 16: Compute $\mathcal{L}$ via L-BFGS with strong Wolfe line search
 - 17: **If** loss stagnates **then break**
 - 18: **end for**
 - 19: **return** θ, α
-

3.3. Experimental setup

All computations are performed in dimensionless form using the scaling of Section 2; results are converted back to physical units where indicated. The physical parameters adopted throughout the study are listed in Table 1.

Reference solution. A finite difference (FD) solver on a uniform spatial mesh with explicit time stepping and a velocity-based tip propagation condition Nordgren (1972) provides the reference solution for accuracy assessment. The FD solver uses the same dimensionless parameters and scaling as the PE-PINN framework.

Table 1

Physical parameters adopted in the numerical experiments.

Parameter	Symbol	Value
Plane-strain Young's modulus	E'	[TBD]
Fracture height	H	[TBD]
Fluid viscosity	μ	[TBD]
Leak-off coefficient	C_L	[TBD]
Injection rate	Q_0	[TBD]
Simulation time	$T_{\max}$	[TBD]

Error metrics. Let $\widehat{W}(x_i, t_i)$ denote the PE-PINN prediction and $W_{\text{ref}}(x_i, t_i)$ the FD reference on a uniform evaluation grid of size N . Accuracy is quantified via the relative L_2 error and the normalized maximum error:

$$\varepsilon_{L_2} = \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (\widehat{W}_i - W_{\text{ref},i})^2}}{\max_i |W_{\text{ref},i}|}, \quad (23)$$

$$\varepsilon_{\max} = \frac{\max_i |\widehat{W}_i - W_{\text{ref},i}|}{\max_i |W_{\text{ref},i}|}. \quad (24)$$

Both metrics are normalized by $\max |W_{\text{ref}}|$ to enable comparison across different time instants and parameter settings.

Ablation variants. To isolate the contribution of each component of the ansatz (12), four model variants are constructed:

- (i) PE-PINN (full): all three enhancements active;
- (ii) No-time-power: $\alpha \equiv 0$ in (12);
- (iii) No-tip-shape: $\Omega_{\text{tip}}(\zeta) \equiv 1$;
- (iv) Plain PINN: $\widehat{W}(x, t) = \mathcal{N}_{\theta}(x, t)$ without any physics-enhanced components.

All variants share the identical network architecture, collocation sets, and training protocol of Algorithm 1.

Sensitivity analysis. The influence of network capacity is characterized by varying the number of hidden layers $D \in \{2, 3, 4, 5, 6\}$ and the number of neurons per layer $W \in \{32, 64, 128, 256, 512\}$, producing a grid of 25 configurations. For each (D, W) pair, the full training procedure of Section 3.2 is executed and the final L-BFGS loss and wall-clock time are recorded.

4. Experiments

4.1. Model validation

The proposed method is evaluated against the finite difference baseline. Fracture width profiles at representative times and the crack length evolution are compared.

4.2. Ablation study

The ablation study quantifies the individual contribution of each component of the proposed framework.

4.3. Sensitivity analysis

The sensitivity analysis characterizes the influence of network depth D and width W on prediction accuracy. Table 2 reports the final L-BFGS loss for 25 configurations spanning $D \in \{2, 3, 4, 5, 6\}$ and $W \in \{32, 64, 128, 256, 512\}$.

The results reveal a clear trend: increasing depth and width generally improves accuracy, but with diminishing returns beyond $D = 4$ and $W = 64$. The optimal configuration is $D = 4$, $W = 64$ with a final loss of 4.71×10^{-3} , achieving near-optimal accuracy (within 2.8% of the global best $D = 6$, $W = 256$) with only 12,738 parameters. Notably, the shallowest network ($D = 2$, $W = 32$) performs an order of magnitude worse, while excessively wide shallow networks (e.g., $D = 2$, $W = 512$) fail to converge effectively, as evidenced by the 3.13×10^{-2} loss. Deeper

Table 2

Comparison of the total loss under different combinations of hidden layers and neurons per layer.

D	$W = 32$	$W = 64$	$W = 128$	$W = 256$	$W = 512$
2	4.41×10^{-2}	9.03×10^{-3}	1.05×10^{-2}	8.92×10^{-3}	3.13×10^{-2}
3	2.50×10^{-2}	5.02×10^{-3}	5.00×10^{-3}	5.58×10^{-3}	6.29×10^{-3}
4	5.17×10^{-3}	4.71×10^{-3}	5.10×10^{-3}	4.72×10^{-3}	4.82×10^{-3}
5	6.37×10^{-3}	4.74×10^{-3}	4.63×10^{-3}	4.69×10^{-3}	4.75×10^{-3}
6	4.77×10^{-3}	4.68×10^{-3}	4.65×10^{-3}	4.58×10^{-3}	4.74×10^{-3}

Table 3

Comparison of the training time under different combinations of hidden layers and neurons per layer.

D	$W = 32$	$W = 64$	$W = 128$	$W = 256$	$W = 512$
2	128	358	340	350	237
3	140	425	543	543	884
4	537	450	520	677	1212
5	361	431	626	944	1330
6	615	596	614	824	1312

networks ($D \geq 4$) exhibit greater robustness, with all configurations achieving losses within a factor of two of the optimum.

Table 3 reports the corresponding training times. The computational cost grows roughly linearly with the number of parameters, with the shallowest network ($D = 2$, $W = 32$) completing in 128 s and the largest ($D = 5$, $W = 512$) requiring 1330 s. The optimal $D = 4$, $W = 64$ configuration trains in 450 s, representing a practical balance between accuracy and computational efficiency. The converged values of the learnable time-power exponent α consistently fall within the physically plausible range of $[0.24, 0.27]$, confirming that the viscosity-dominated regime ($\alpha \approx 0.25$) is correctly identified across all configurations.

5. Conclusion

This paper presented PE-PINN, a Physics-Enhanced PINN framework for the PKN model of hydraulic fracture propagation. The proposed method addresses three fundamental challenges of applying PINNs to moving boundary problems: the transient tip shape function analytically encodes the transition between asymptotic regimes, the causal arrival-time constraint restricts the PDE residual to the physically valid region, and the learnable time-power exponent adaptively captures the temporal scaling behavior. Numerical experiments demonstrate competitive accuracy against a conventional finite difference solver while maintaining a mesh-free formulation. Future work will extend the framework to more complex fracture propagation scenarios.

Acknowledgments

This work was supported by [FUNDING INFORMATION TO BE ADDED].

CRedit authorship contribution statement

[**Author 1**]: Conceptualization, Methodology, Software, Writing. [**Author 2**]: Validation, Writing - Review & Editing.

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